Workload Protection in Azure Local

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Agenda

- **Objective:** Review Microsoft recommended application protection and backup mechanisms in Azure Local
- **Scope:** Implementing Windows Defender Application Control (WDAC) and Microsoft Azure Backup Server (MABS) for protecting Azure Local workloads

Application Control in Azure Local

- Purpose:
  - WDAC reduces **attack surface** using **policy-based allow/deny** for apps and drivers.
- Visibility and management:
  - **WDAC settings** are visible in the **Azure portal** once the **Microsoft Cloud Security Benchmark (MCSB)** is applied
  - **Management** is performed via **PowerShell**.

WDAC Policy Modes

- Modes:
  - **Audit** (log only) and **Enforced** (block untrusted)
  - Configurable per node
  - Reported by `Get-AsWdacPolicyMode`
- Scope of control:
  - `Enable-AsWdacPolicy` (all cluster nodes)
  - `Enable-ASLocalWDACPolicy` (local node)
  - Depends on **rollout strategy**
- Recommended Pattern:
  - Temporarily set **Audit** to evaluate **new software/agents**,
  - Then set to **Enforced** with a **supplemental policy** in place.

Switching WDAC Modes

- Check current mode per node:
  - `Get-AsWdacPolicyMode`
- Switch cluster mode:
  - `Enable-AsWdacPolicy -Mode Audit # or Enforced`
- Orchestrator propagation takes ~2–3 minutes
  - Re-check with `Get-AsWdacPolicyMode`

Troubleshooting and Remediation

- **Blocked apps** create **Event Log** entries indicating **policy reason**.
  - Use the **Application Control operational guide** to interpret and remediate.
- Combine **publisher rules** with **hash fallback** to safely allow required binaries without over-permitting.

Supplemental Policy for Binaries

- In **Enforced** mode, allow designated software via a **supplemental WDAC policy**.
- **Rollout flow**:
  - Switch to Audit → Install/Test → Create supplemental policy → Return to Enforced.

Authoring & Deploying Supplemental Policies

- Create policy (publisher level + hash fallback) scanning only the agent folder:

```
New-CIPolicy -MultiplePolicyFormat -Level Publisher `
-FilePath c:\wdac\Contoso-policy.xml -UserPEs -Fallback Hash `
-ScanPath c:\software\codetoscan
```

- Set metadata:

```
Set-CIPolicyVersion -FilePath c:\wdac\Contoso-policy.xml -Version 1.0.0.1
Set-CIPolicyIdInfo -FilePath c:\wdac\Contoso-policy.xml `
-PolicyID "Contoso-Policy_1.0.0.1" -PolicyName "Contoso-Policy“
```

- Deploy & verify:

```
Add-ASWDACSupplementalPolicy -Path c:\wdac\Contoso-policy.xml
Get-ASLocalWDACPolicyInfo
```

WDAC and MABS & ASR Deployments

- **Default WDAC** can **block agent deployment** or **recovery scripts**
  - **Audit mode** is the recommended staging mode for installation and testing.
- After validation, **return to Enforced** with a **supplemental policy** so the agents and scripts run reliably under strict controls

Resiliency Fundamentals on Azure Local

- By default, Azure Local VMs are designed for **high availability**
  - On **node failure**, VMs **restart** on remaining nodes (subject to capacity).
- **Well-architected workloads** pair platform HA with **backup strategies** and **data replication** to withstand corruption, deletion, and site incidents.

Introducing Microsoft Azure Backup Server (MABS)

- **MABS** delivers **hybrid backup**:
  - **short-term retention** on local disk
  - **long-term retention** in Recovery Services vault via Azure Backup
- Protect **Azure Local VMs** at **host-level** and/or **guest-level** to optimize **RPO/RTO**

Host-Level Backup Characteristics

- Agent installed on each Azure Local host
  - Captures **entire VM + virtual disks**
  - **OS-agnostic** at guest layer.
- Enables full VM restores to same or alternate cluster
  - Alternate restores produce **unmanaged VMs**
  - Not **application-aware** beyond **VSS**.

Guest-Level Backup Characteristics

- **Agent inside VM** for **application-consistent** backups (for **VSS-aware apps**).
  - Supports **item-level** restores (e.g., **databases**, **files**).
- **Management overhead is higher**
  - Full VM rebuild typically precedes restoring data **inside** the guest.

Blended Strategy: Both Host- and Guest-Level

- **Guest-level** for **application-aware**, **item-level** recovery in critical systems.
- **Host-level** for **quick VM** recovery from infra events.
- Many orgs **combine** the two to align **RPO/RTO** across tiers.

Supported Scenarios & Limitations

- **Host protection** covers **System State/BMR**
  - **Agents on host** required.
- **Clustered VMs (CSV)** supported with **agent on each node**
  - **VM move within a cluster** continues protection
  - **Move to different cluster** is **not supported**.
- **Alternate Location Recovery (ALR)** restores as **unmanaged VM**

Capacity Planning & Pre-Flight Requirements

- **Example sizing:**
  - **800 VMs × 100 GB ≈ 80 TB** required for **backup storage** per **MABS** server.
- **Hyper-V role** on MABS server is required for **item-level recovery**
- Consider **domain/auth** model (trusted/child vs workgroup/untrusted).

Firewall & WDAC Caveats During Agent Deployment

- If **Azure Benefits** is enabled
  - disable firewall rule `AzsHci-ImdsAttestation-Block-TCP-In` to allow Agent WMI Queries.
- **Default Application Control** may block agent installs
  - Switch WDAC to **Audit** before installation
  - Return to **Enforced** after and rely on supplemental policy.

Creating Protection Groups, Schedules & Retention

- In MABS Administrator Console:
  - **Create Protection Group** → **Select VMs** (cluster HA vs unmanaged on nodes) → set **short-term disk** and optional **online protection** to Azure.
  - **Configure synchronization frequency, retention range**, and whether to run **express full** just before recovery points.

Recovery Options

- Recover to original instance:
  - Deletes original VHD + checkpoints and restores via **Hyper-V VSS**
  - Retains Azure management when in-place.
- Recover as VM to any host (ALR):
  - Restores as **unmanaged VM**
- Copy to network folder:
  - **Item-Level Recovery (ILR)** restores **files, folders, VHDs** without an agent inside the guest.

Network Throttling & Consistency Checks

- **Enable Network bandwidth throttling** in limited bandwidth scenarios
- Automate **consistency checks** on **inconsistency** or **schedule**
  - run **manual** checks when needed.

Backup Frequency, Retention Tiers & Restore Testing

- Align **backup frequency** to **RPO**
  - Use **incrementals** nightly or multiple times/day based on business criticality.
- Define **retention tiers** matching compliance requirements
  - (e.g., **daily/2 weeks**, **monthly/6 months**, **yearly/7 years**)
- **Regularly test restores**
  - full VM and data item recovery in an **isolated network or lab cluster**.

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